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ENTERPRISE SURVEY DATA AND THE ETR CORE AI MODEL

From Enterprise AI Adoption to Semicap Equipment: Sizing the AI Build-out



SEMICON TAIWAN 2026 | MARKET AND INDUSTRY TREND FORUM

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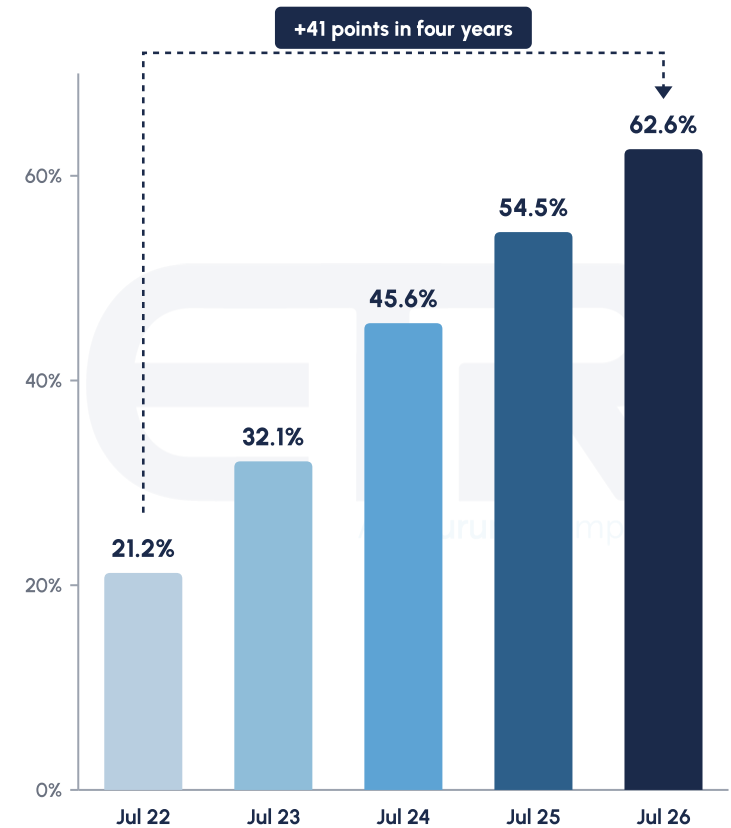
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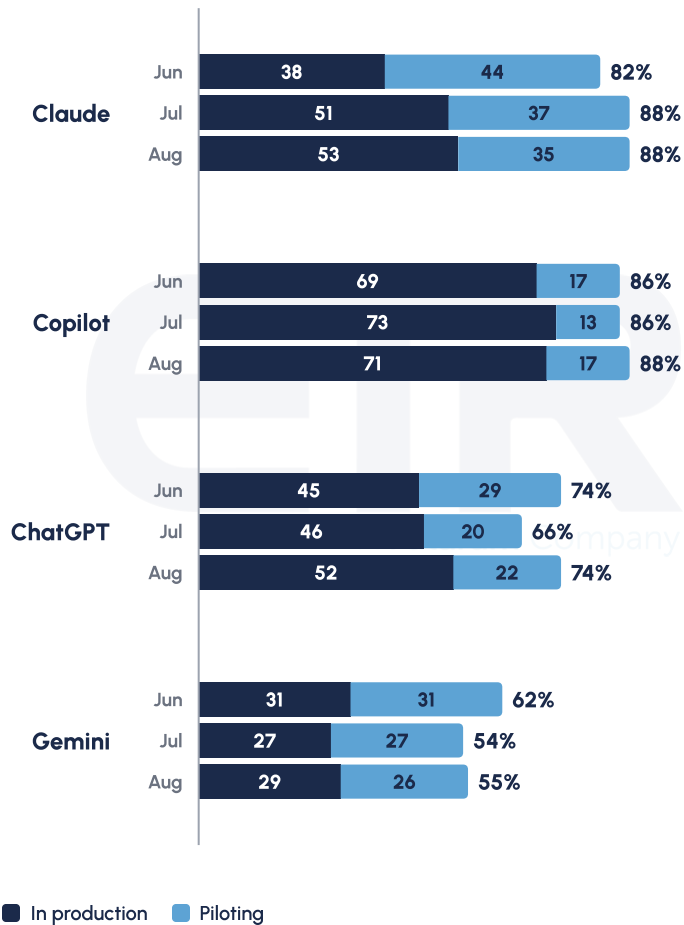
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AI adoption is near-universal, and demand spans multiple vendors.

ML / AI PERVASION OF THE TSIS (%)



BUSINESS RELATIONSHIP BY AI TOOL (%)



■ In production ■ Piloting

Pervasion = non-replacing sector citations as a share of the survey.

Source: ETR Technology Spending Intentions Survey, July 2026 (N=1,636; 1,024 citing ML/AI); ETR AI Tools Pulse, June to August 2026 (N=100 each).

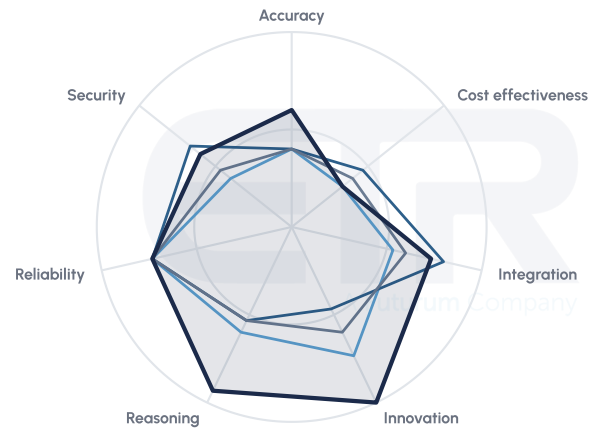
OUR TAKE ON THE DATA

- **ML / AI adoption is increasing:** pervasion amongst 1,636 enterprises surveyed tripled in four years to **62.6%**.
- **The largest buyers are ahead of the curve:** Fortune 100 pervasion is **71%**, up 10 points over the year.
- **No winner takes all dynamic:** including pilots, Claude and Copilot are in use or evaluated by **88%** of enterprises.
- **Room for higher penetration:** Claude is piloted by **35%** of enterprises vs. 17% for Copilot, with more of its base still to convert.

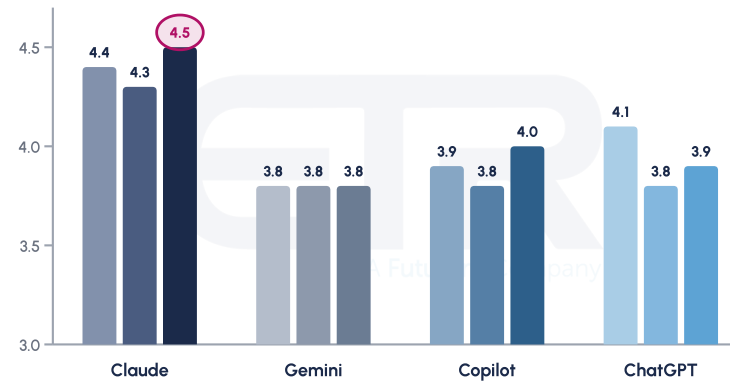
AI use at enterprises will be ubiquitous before the end of the decade.

Claude leads on most metrics, but cost is affecting user perception.

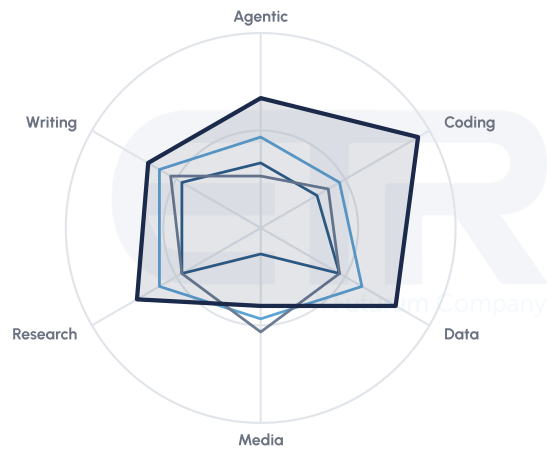
CAPABILITY RATINGS, AUG 2026 (1-5)



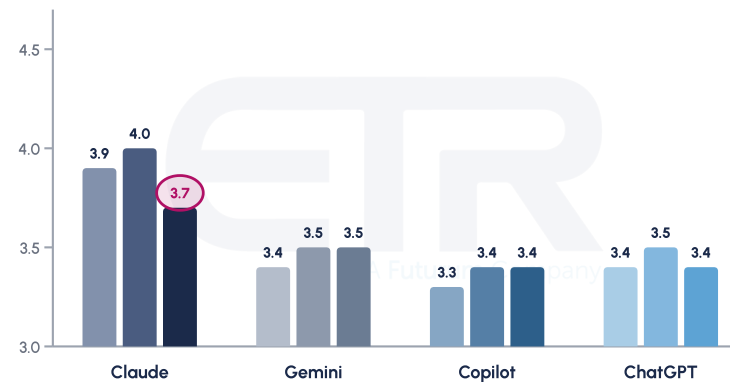
OVERALL SATISFACTION, JUN TO AUG (1-5)



USE-CASE PERFORMANCE, AUG 2026 (1-5)



30-DAY PERCEPTION, JUN TO AUG (1-5)



■ Claude ■ Gemini ■ Copilot ■ ChatGPT

OUR TAKE ON THE DATA

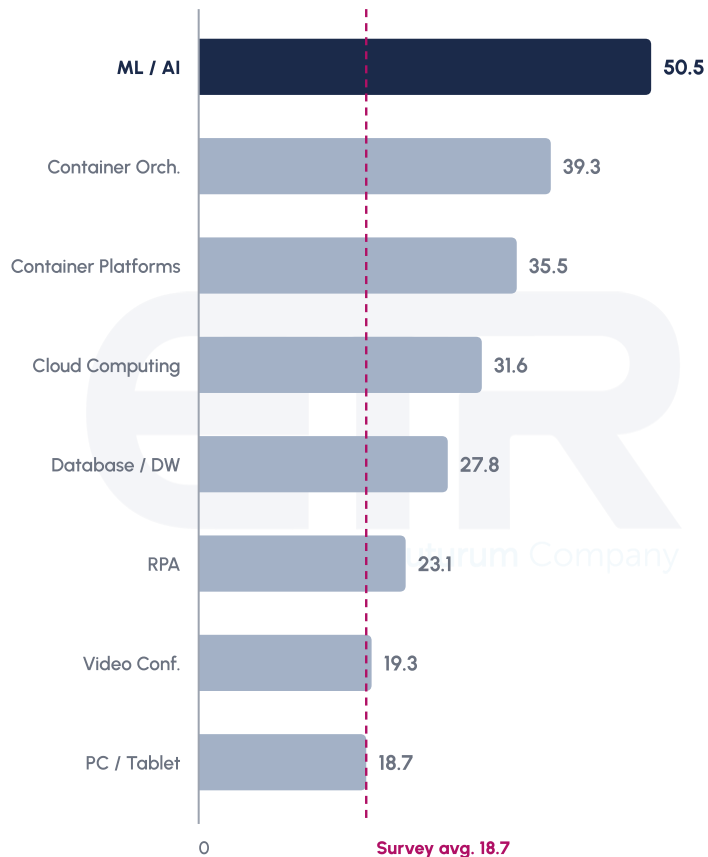
- **Claude leads on capability:** it is well ahead of peers on accuracy, pace of innovation (**4.5**) and reasoning (4.4).
- **Claude is the preferred tool for 5/6 use cases:** coding, data, and agentic performance, but also research and writing.
- **Satisfaction is at a high:** Claude reached **4.5** in our August survey. Peers rank between 3.8 and 4.0.
- **Cost is increasingly perceived to be an issue:** Claude's perception (change in sentiment) slipped from 4.0 to **3.7**, and it ranks below peers on cost effectiveness (3.5).

Does escalating cost affect the pace of adoption?

Source: ETR AI Tools Pulse, August 2026 (N=100). Capability, use-case, satisfaction and perception ratings; deltas vs. July 2026.

AI leads every sector on intent, metered usage keeps increasing, but hypergrowth may tail off.

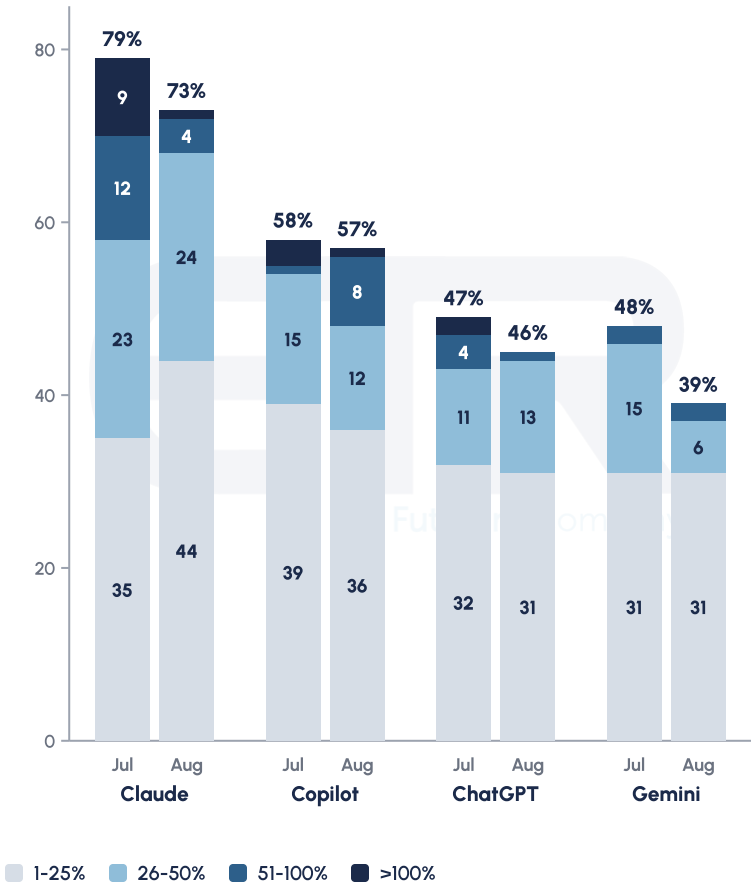
NET SCORE BY SECTOR, JUL 2026 (TOP 8 OF 29)



Net Score = adoption plus increase, less decrease and replacing.

Source: ETR Technology Spending Intentions Survey, July 2026 (N=1,024); ETR AI Tools Pulse, July and August 2026 (N=100 each).

INCREASE IN METERED CONSUMPTION, PAST 30 DAYS (%)



Base: accounts with metered tracking. Segments may not sum to the total because of rounding.

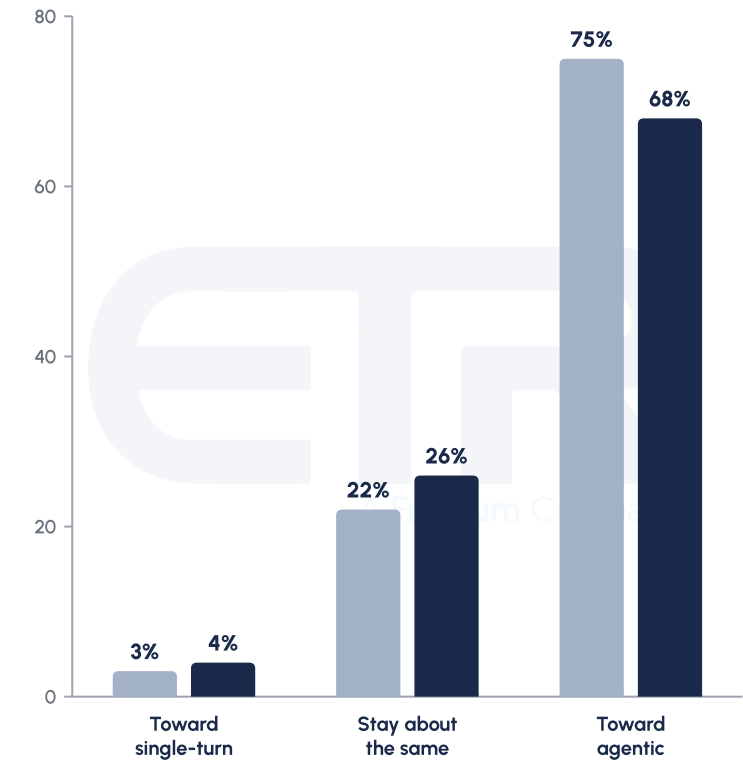
OUR TAKE ON THE DATA

- **AI leads every sector on intent:** a Net Score of **50.5** (adoption plus increased spending) tops all 29 sectors we track.
- **Usage is still broadening:** **73%** of Claude accounts report higher month-on-month consumption in August, ahead of Copilot at 57% and ChatGPT at 46%.
- **Early signs of a normalization in consumption?** Accounts that grew consumption over 50% month-on-month dropped from 21% in July to **5%** in August.

What drives higher enterprise token consumption?

Enterprises are shifting to agentic usage, with more tasks the largest growth driver.

EXPECTED CHANGE IN USAGE MIX, NEXT 3 MONTHS (%)

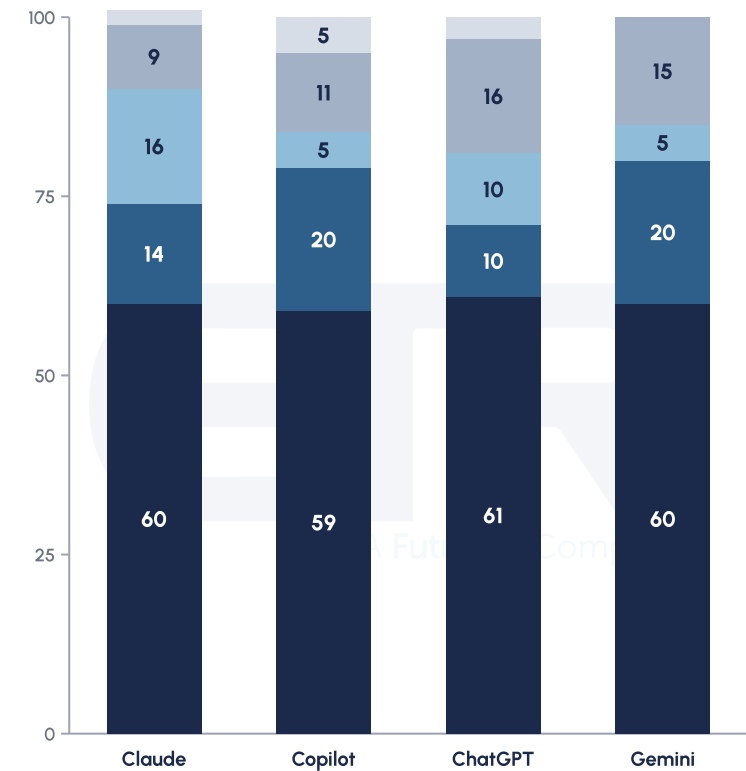


■ July 2026 ■ August 2026

Toward agentic combines modestly (55%) and significantly (13%) in August.

Source: ETR AI Tools Pulse, July and August 2026 (N=100 each).

LARGEST DRIVER OF CONSUMPTION GROWTH, AUG 2026 (%)



■ More tasks ■ More tokens per task
 ■ Switched to pricier model ■ Vendor raised price ■ Unsure

Base: accounts reporting increased consumption in August. Gemini N=20.

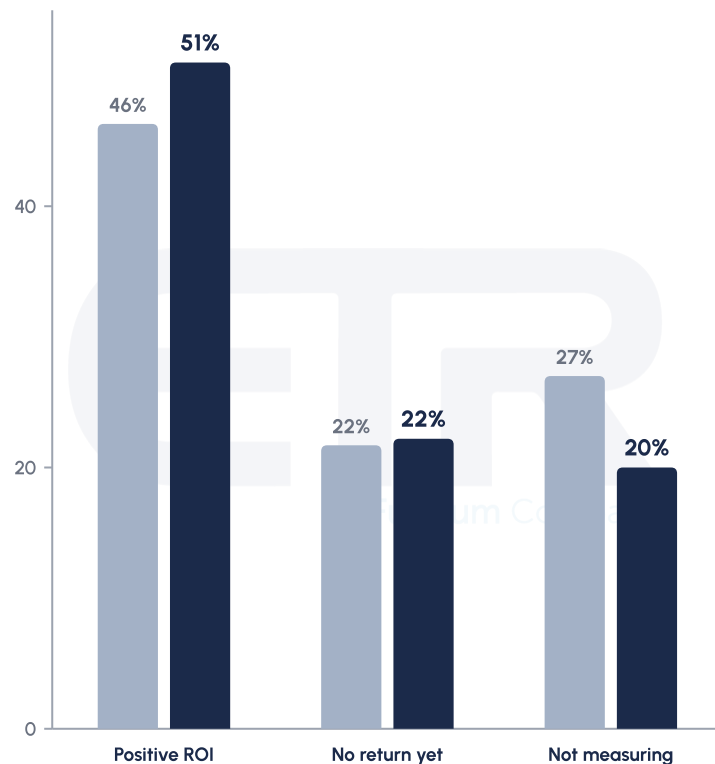
OUR TAKE ON THE DATA

- **Agentic AI on the rise:** 68% of enterprises expect a further shift toward agentic usage over the next three months.
- **Growth is driven by more tasks:** around 60% name a higher number of tasks as the largest driver for growth, with consistent ratings across all tools.
- **Increasing complexity is a secondary driver:** 10% to 20% cite more tokens per task as the largest driver of higher consumption.

Is higher consumption resulting in returns on investment and how is growth funded?

The ROI on AI is positive for most enterprises, with funding mostly new budget but also reallocations.

REALIZED ROI FROM AI SPENDING (%)

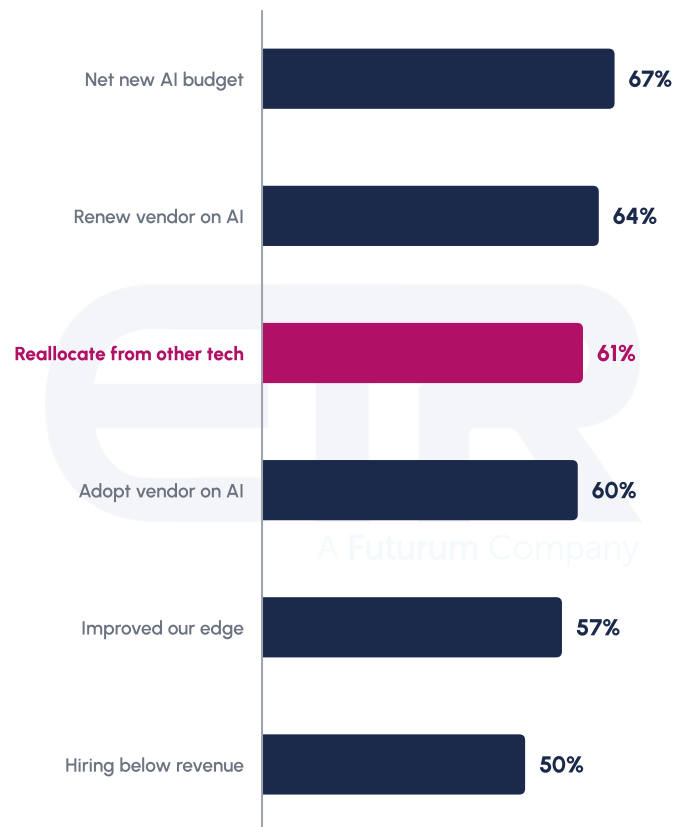


■ November 2024 ■ July 2026

ETR AI Product Series, N=600 each period.

Source: ETR AI Product Series, July 2026 (N=600); ROI compared with November 2024 (N=600).

AGREE OR STRONGLY AGREE WITH THE STATEMENT (%)



Statements posed to IT decision makers on AI budget and vendor choice.

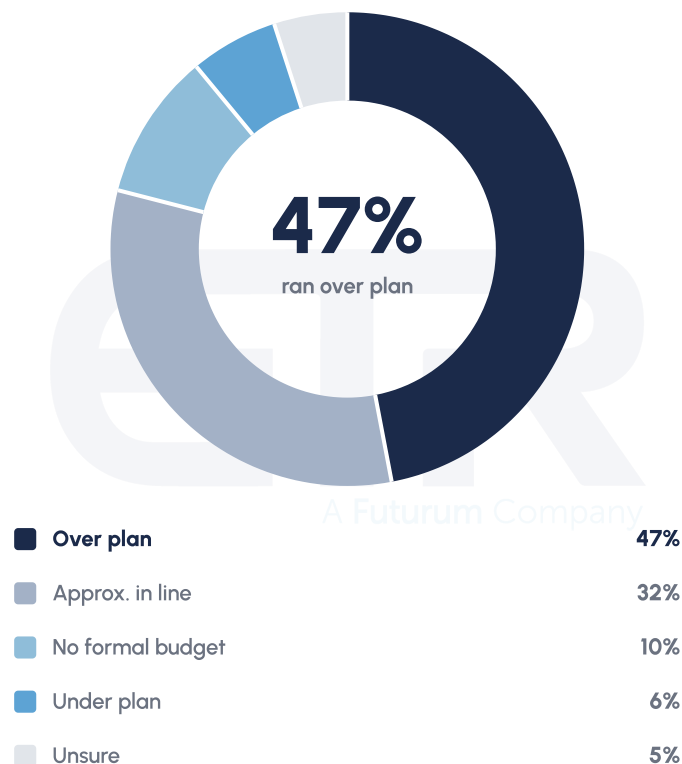
OUR TAKE ON THE DATA

- **AI returns are improving:** positive ROI of **51%** in July, the highest of eight surveys, up from 46% in late 2024.
- **Investments are rationalizing:** the share of enterprises not measuring ROI came down from 27% to **20%**.
- **New budget is being allocated:** **67%** of companies have new AI budget available next year.
- **AI is prioritized in vendor selection:** AI features drive **64%** of renewals and 60% of new deals.
- **Budget is cut from other tech areas:** **61%** of enterprises reallocate budget from other areas.

Is AI spending tracking in line with expectations, and where is budget reallocated from?

AI budgets are overrunning at close to half of enterprises; consultants and hardware refresh are funding the gap.

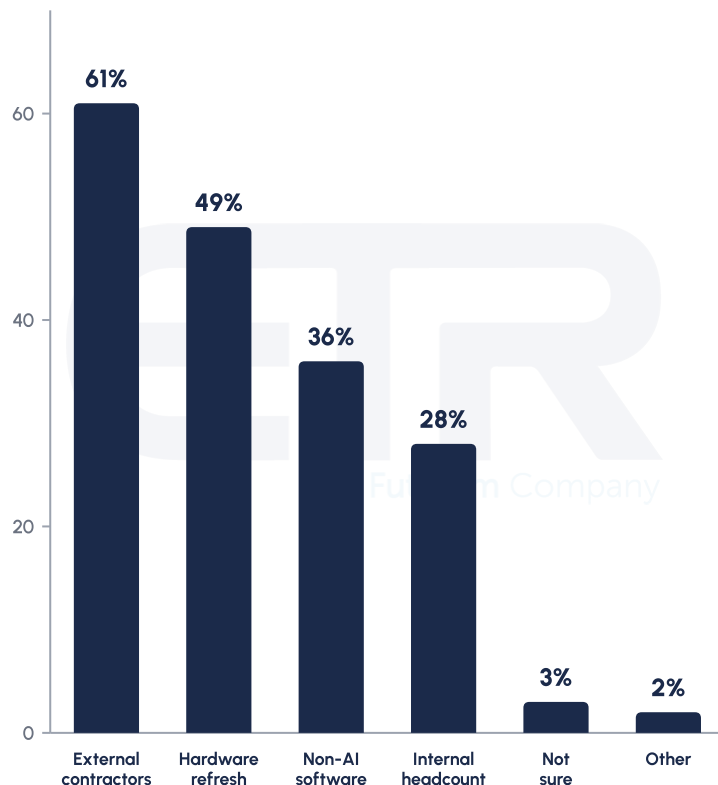
ACTUAL AI SPEND VS. PLANNED BUDGET (%)



N=1,636 IT decision-makers. Balance unsure.

Source: ETR Macro Views, July 2026 (N=1,636). Over-plan N=767; reallocation N=297.

WHERE BUDGET IS REALLOCATED FROM (%)



N=297 who reallocate IT budget to cover AI overruns. Multi-select.

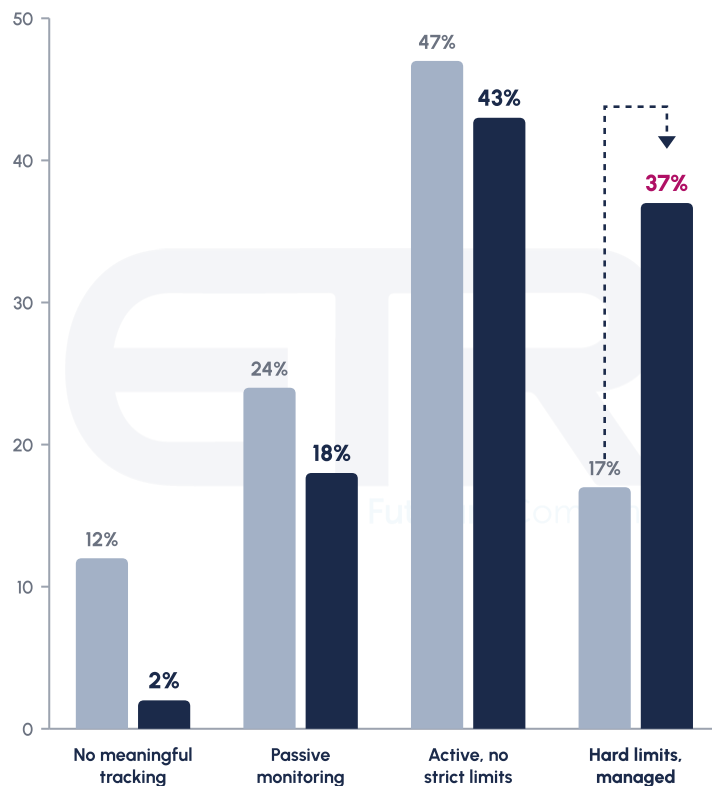
OUR TAKE ON THE DATA

- **Half the market is over budget:** 47% run above plan, 32% in line, and only 6% came in under budget.
- **Enterprises don't slow down:** 48% seek extra budget and 43% reconcile budgets next cycle. Only 17% pause or scale back AI deployments.
- **Contractors are cut first:** external contractors and consultants absorb 61% of reallocations.
- **Hardware refresh is delayed:** 49% of companies push out refresh cycles, followed by non-AI software and headcount.

How do companies plan to control token over-consumption?

Agentic ambitions may run into budget reality.

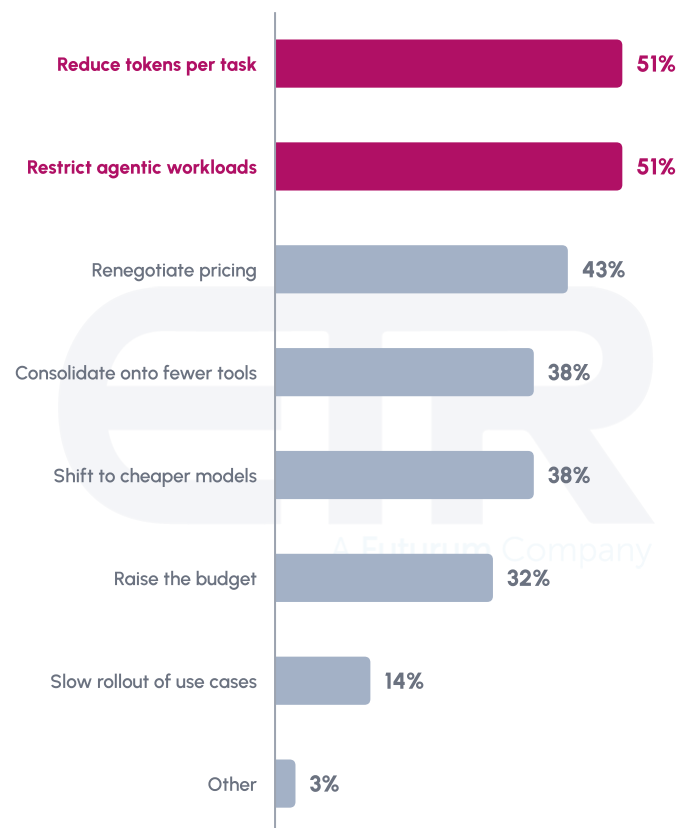
HOW AI CONSUMPTION IS GOVERNED (%)



■ Today ■ Expected in 3 months

August 2026, N=100.

METHODS USED TO STAY INSIDE LIMITS (%)



Base: respondents managing or planning to manage against hard limits (N=37).

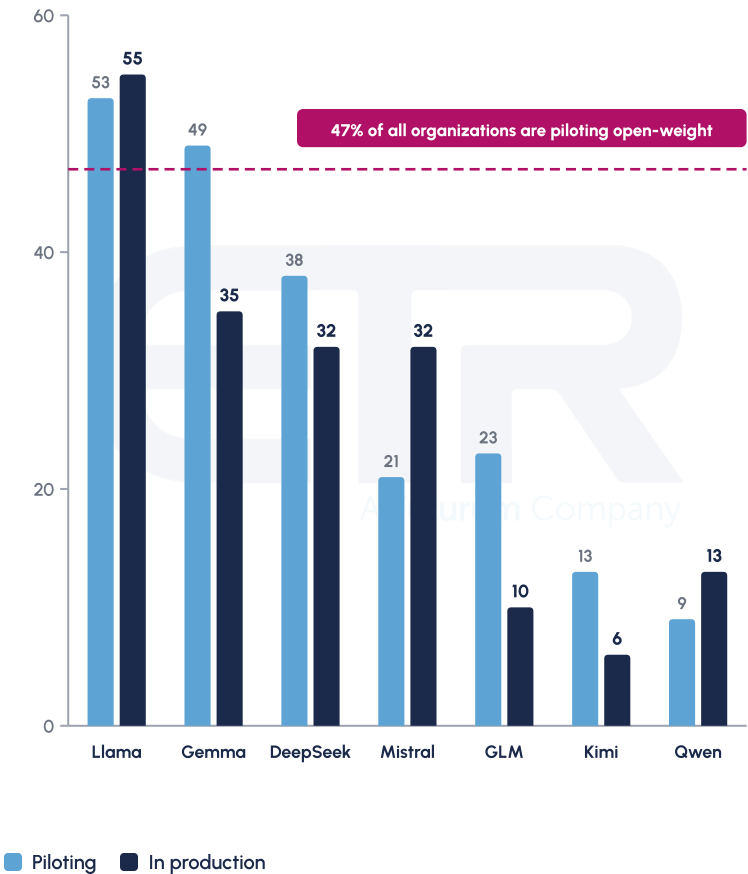
OUR TAKE ON THE DATA

- **Hard limits are still rare:** only **17%** manage consumption against hard limits today.
- **But ambitions are there:** **37%** of enterprises expect to have hard limits within three months.
- **Curtailing agentic work is a key cost lever:** restricting agentic workloads and reducing tokens per task are the joint top methods to control budgets, at **51%** each.

Budget-driven usage caps may slow agentic adoption in coming years.

Open-weight models are a potential means to reduce costs.

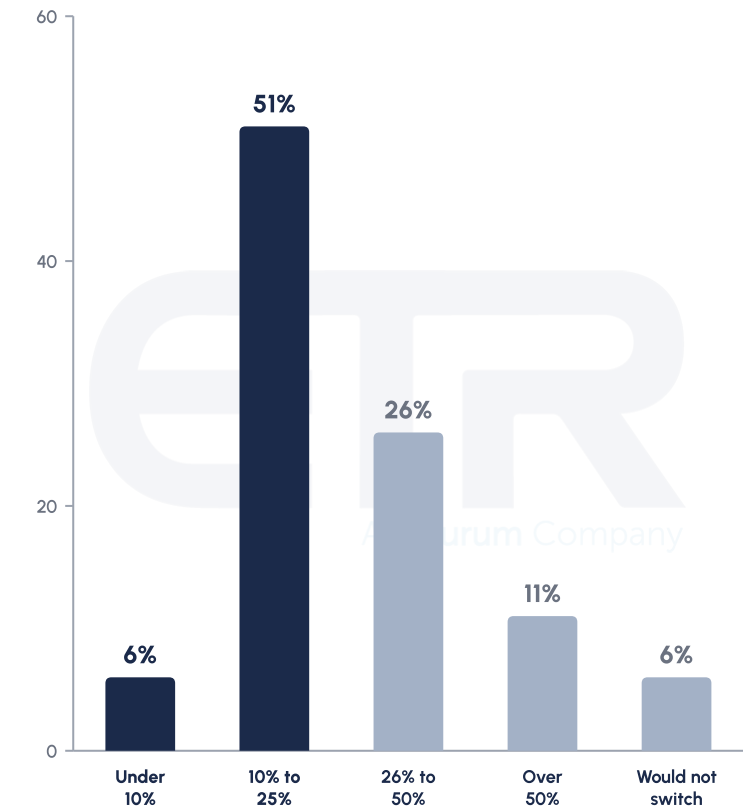
OPEN-WEIGHT FAMILIES IN USE (% OF COHORT)



Family shares are of their own cohort; the dotted line is the share of all organizations.

Source: ETR Open-Weight AI Models Drill Down, July 2026 (N=100; pilots N=47, production N=31).

DISCOUNT NEEDED TO MOVE WORKLOADS TO OPEN SOURCE (%)



Pilots (N=47), at comparable performance. Sums to 100%.

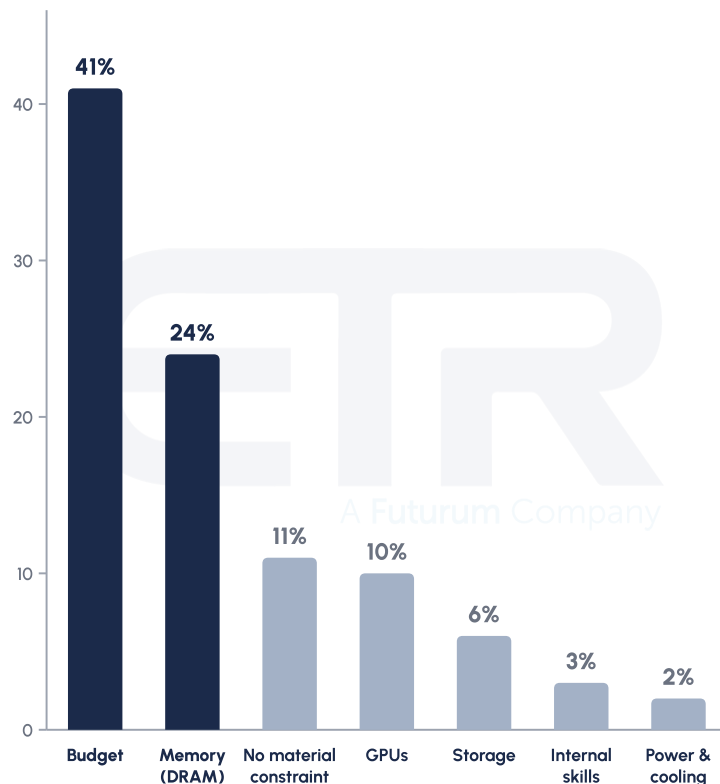
OUR TAKE ON THE DATA

- **Open weights are already in production: 47%** of organizations are piloting and 31% run them in production.
- **Llama leads, but China is also a major contributor:** Llama tops pilot and production uptake, but DeepSeek, GLM and Qwen are also frequently used.
- **Small discounts can move workloads:** at parity in performance, **57%** of pilots would switch for a discount of 25% or less.
- **This is even more pronounced for production workloads:** among those already in production, **74%** would switch for 25% or less.

Open-source adoption is one of the key trends to track in the second half of the year.

Hardware prices, especially of memory, are a major source of concern for enterprises.

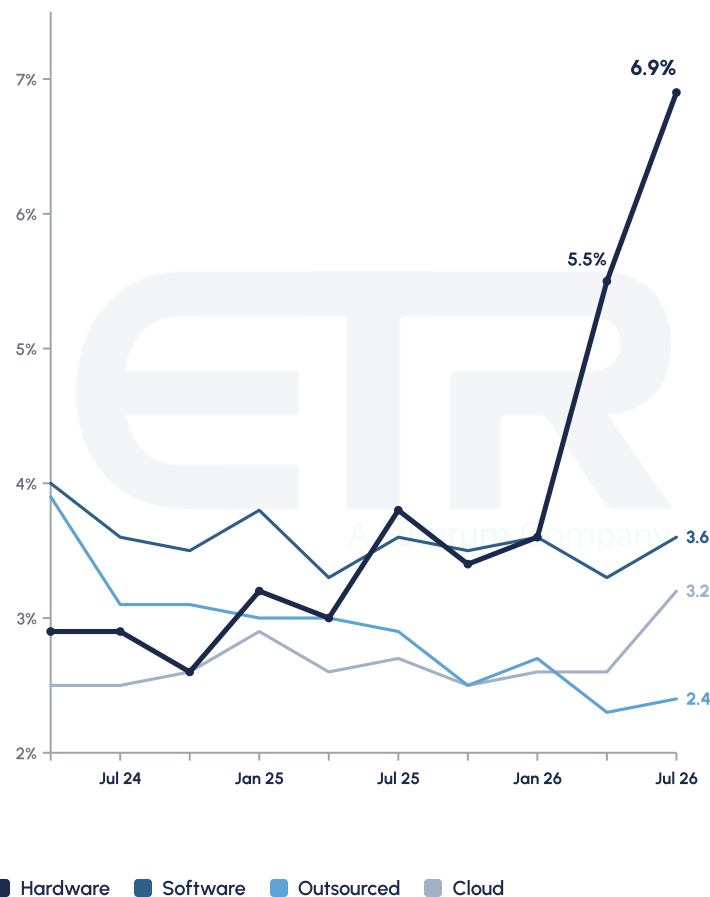
BIGGEST CONSTRAINT ON DEPLOYING MORE COMPUTE (%)



Single-select. N=103 IT leaders with hardware purview.

Source: ETR Hardware Purchasing Drill Down, July 2026 (N=103); ETR Macro Views, July 2026 (N=1,636).

REPORTED PRICE CHANGE VS. 7-12 MONTHS AGO (%)



Macro Views, July 2026 (N=1,636). Hardware covers all purchasing categories, AI parts included.

OUR TAKE ON THE DATA

- **Budget is the main constraint, followed by memory:** 41% of buyers name budget as the #1 constraint in deploying more compute; memory is #2, well ahead of GPUs.
- **Hardware inflation shows up in enterprise reporting:** hardware pricing up **6.9%** half-on-half in July, from 3.6% in January, while cloud and software were stable in a 2 to 4% band.
- **Memory is ahead of other segments:** 80% of memory buyers report price increases, and 26% of those saw 50% or more.

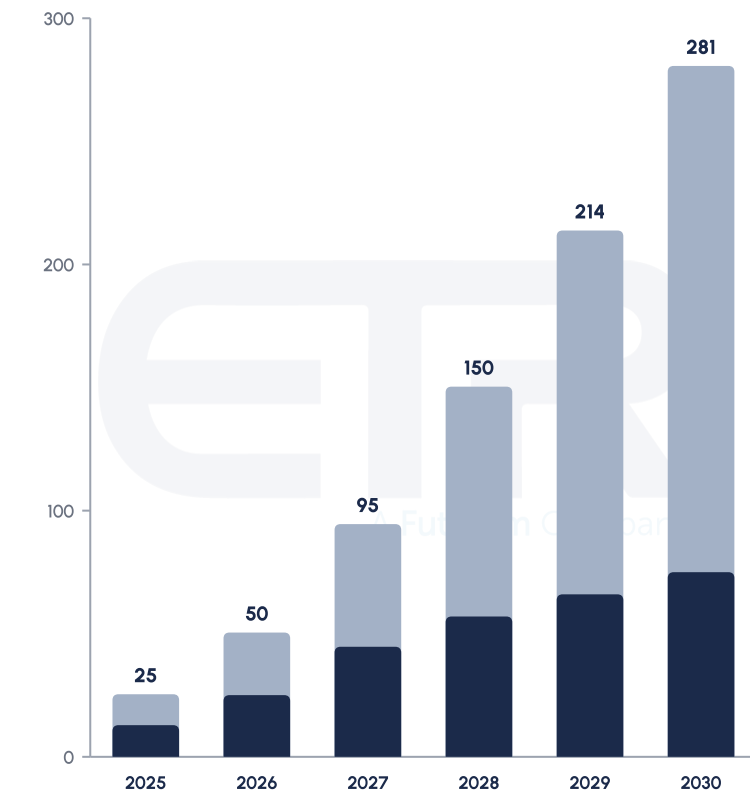
What visibility do we have on the ramp of AI deployments in coming years?

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AI data centers can ramp to over 280 GW of installed capacity by 2030.

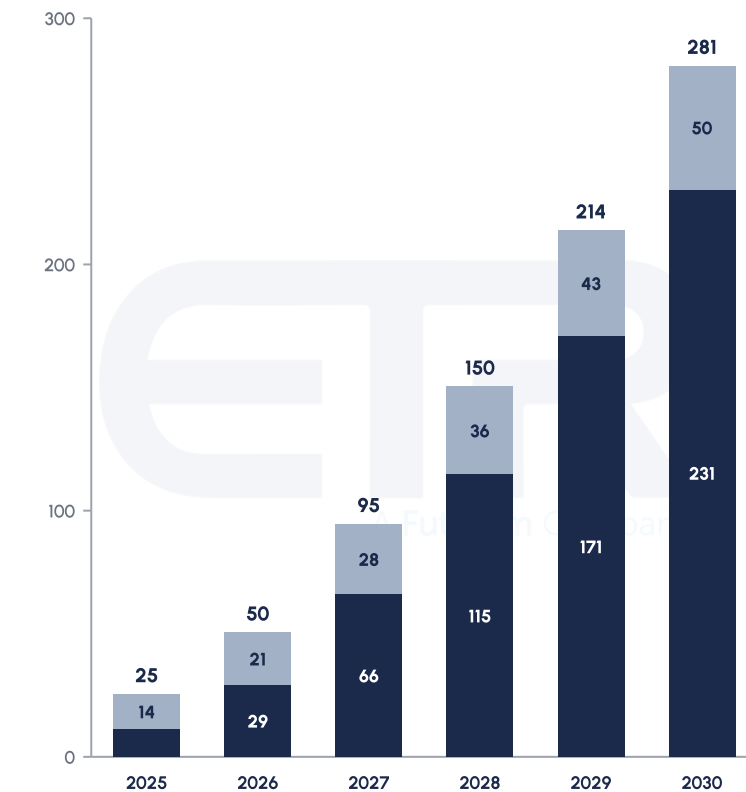
GLOBAL DATACENTER AI CAPACITY (GW)

INSTALLED AI CAPACITY BY WORKLOAD (GW)



■ Installed base ■ Deployed that year

Deployments are the capacity added in that year.



■ Inference ■ Training

Segments may not sum to the total because of rounding.

Source: ETR Core AI Model. Global datacenter AI capacity, deployed and installed; installed base split by workload.

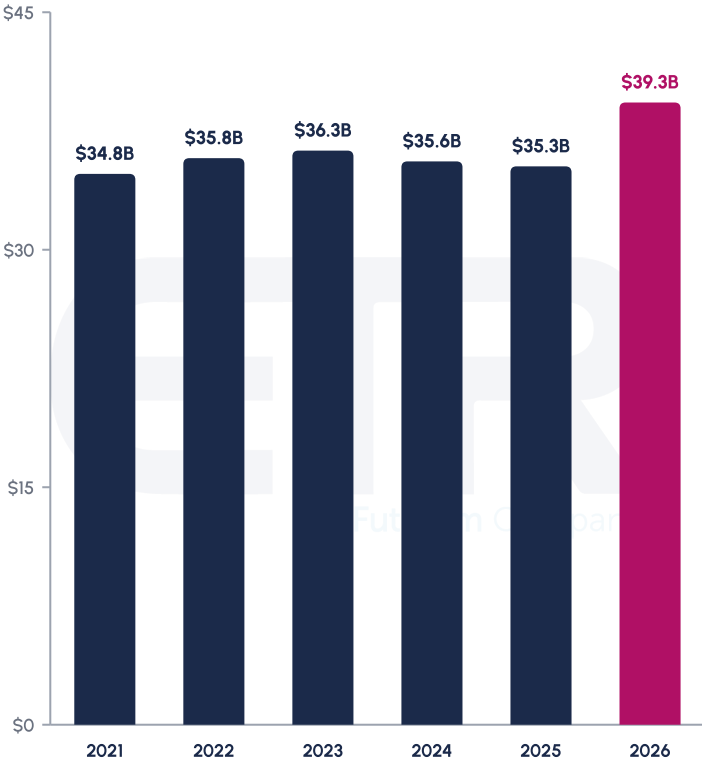
OUR TAKE ON THE DATA

- **Supply chain is positioning for 280 GW by 2030:** more than **11x** the 2025 installed base of 25 GW.
- **Annual deployments reach 75 GW:** up from 13 GW in 2025, a **5.8x** increase across the period.
- **Inference drives most capacity additions:** training more than triples in absolute terms, but declines from **55%** of installed capacity in 2025 to under 20% in 2030.

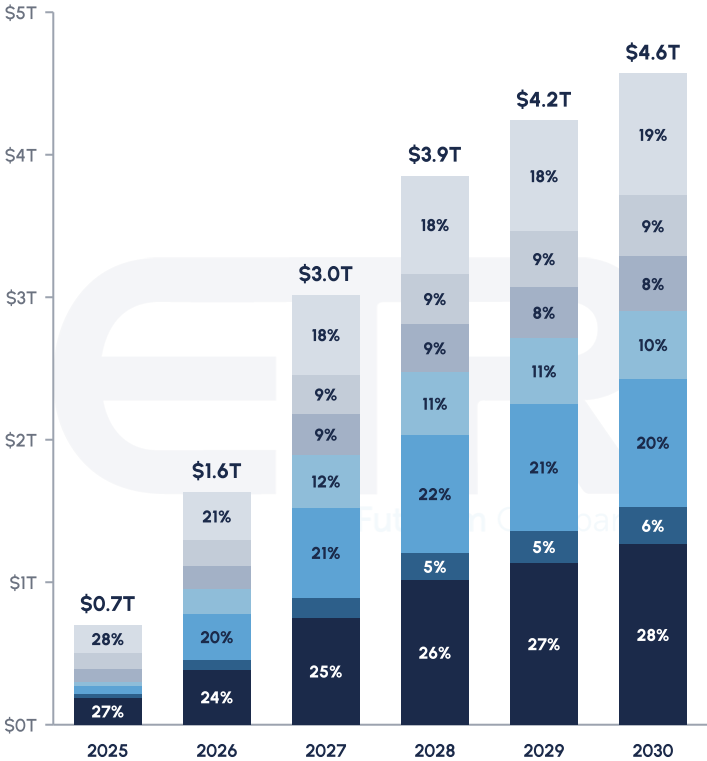
What does it cost to deploy 280 GW of AI capacity?

Capex per GW on the way up; total datacenter capex can exceed \$4T in 2030.

AI DATACENTER CAPEX PER GW (\$B/GW)



TOTAL DATACENTER CAPEX BY CATEGORY (\$T)



■ Accelerators ■ CPUs ■ System DRAM ■ System NAND
■ Networking ■ Other server ■ Physical infrastructure

Total datacenter capex, AI and general purpose. Shares may not sum to 100 because of rounding.

Capex divided by gigawatts deployed in that year.

Source: ETR Core AI Model. AI datacenter capex per GW deployed; total datacenter capex by category.

OUR TAKE ON THE DATA

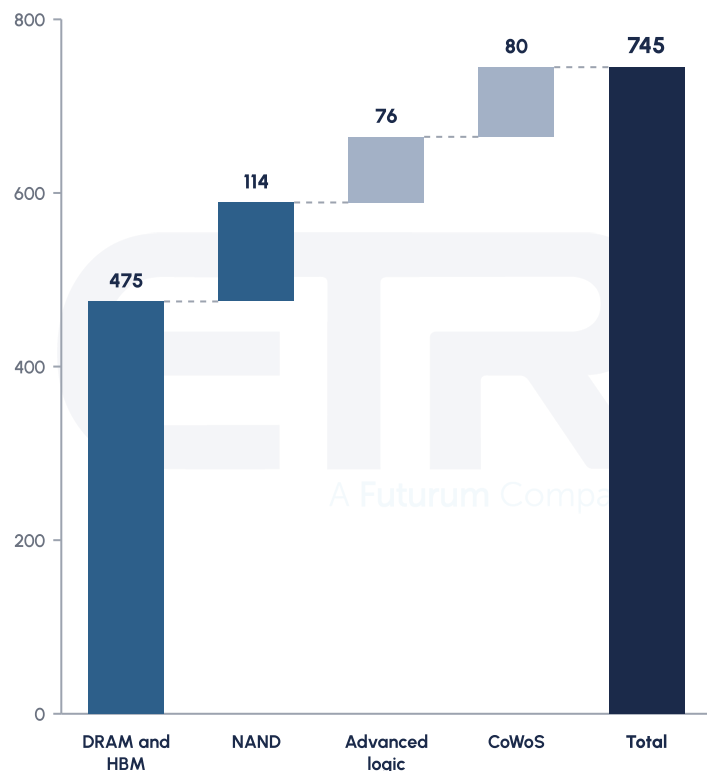
- **Capex per gigawatt is set to increase:** stable at \$35B to \$36B in prior years, but set to increase over **10%** this year, driven by higher component costs.
- **Memory repricing:** system DRAM goes from 7% of datacenter capex in 2025 to **20%** in 2026, and is unlikely to come down materially.
- **CPUs are a smaller moving part:** set to increase from \$64B this year to **over \$250B** in 2030, driven primarily by standalone deployments.
- **Total capex can exceed \$4T in 2030: 2.8x** 2026 spending, with close to \$1.3T of it on accelerators.

How does a GW of capacity translate into silicon and wafer fab equipment demand?

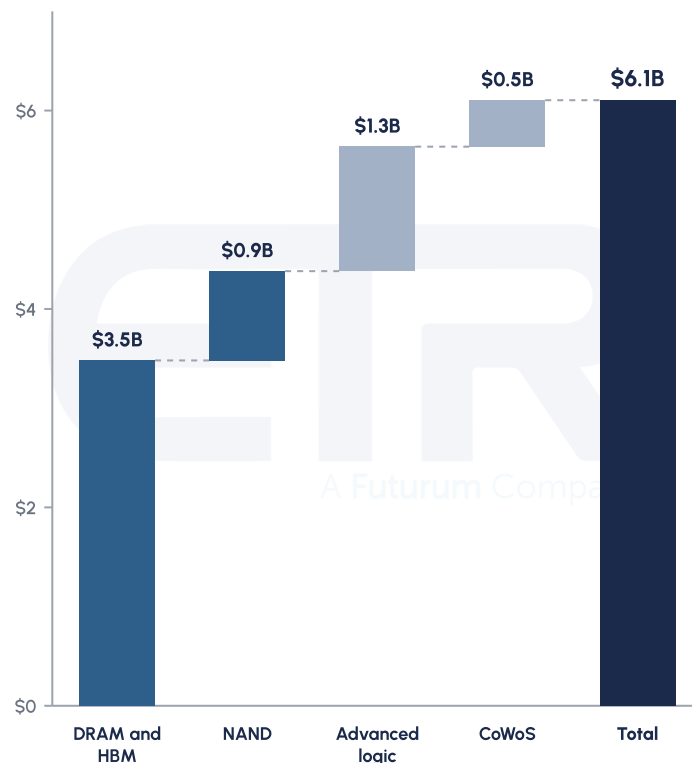
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A GW of capacity requires close to 750k wafers and over \$6B of upfront WFE investment.

WAFERS REQUIRED PER 1 GW DEPLOYED, 2025 (THOUSANDS)



WFE REQUIRED PER 1 GW DEPLOYED, 2025 (\$B)



■ Memory ■ Logic and packaging ■ Total

Memory is 79% of the wafers behind each gigawatt.

Source: ETR Core AI Model. Wafers and wafer fab equipment required per 1 GW of AI capacity deployed in 2025.

OUR TAKE ON THE DATA

- **A gigawatt needs 745,000 wafers of leading-edge silicon on average:** across the GPU, server rack, and related networking.
- **Memory dominates the build-up:** HBM, DRAM and NAND take **over 75%** of the wafers per GW.
- **\$6B of WFE required per GW:** of which DRAM and HBM **\$3.5B**, logic **\$1.3B**, with NAND and CoWoS the remainder.

How does this translate into total WFE spending required for the AI infrastructure build-out?

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AI needs \$570B of cumulative WFE spending by 2030.

AI GW DEPLOYMENT

13 → 75 GW

2025 to 2030E, base case

CUMULATIVE WFE REQUIRED

\$573B

2030E, to install AI capacity

WFE PER GW

\$6.1B → \$7.6B

rising capital intensity

CATEGORY	CAPACITY REQUIRED	CAPITAL INTENSITY	WFE REQUIRED	SHARE OF ADDED SPEND
	KWSPM 2025 → 2030E	\$B per 100 KWSPM 2025 → 2030E	\$B 2025 → 2030E	
● Logic	81K → 552K	\$20B → \$27B	\$16B → \$149B	27%
● HBM	375K → 1,025K	\$9B → \$14B	\$33B → \$143B	22%
● DRAM	134K → 1,340K	\$8B → \$13B	\$11B → \$169B	32%
● NAND	122K → 491K	\$9B → \$13B	\$12B → \$65B	11%
● CoWoS	86K → 508K	\$7B → \$9B	\$6B → \$46B	8%
Total			\$79B → \$573B	100%

OUR TAKE ON THE DATA

- **Capacity requirement:** 75 GW requires over **550k WSPM** of advanced logic, over 2m of DRAM and HBM, and around 500k each of NAND and CoWoS.
- **WFE per GW to increase from \$6.1B to \$7.6B:** backside power and new materials in logic, more EUV layers in DRAM, and higher layer counts in NAND.
- **Cumulative spending required: \$573B** of WFE for AI servers, excluding standalone CPU deployments and related storage.

How does this translate into annual WFE spending over the next 5 years?

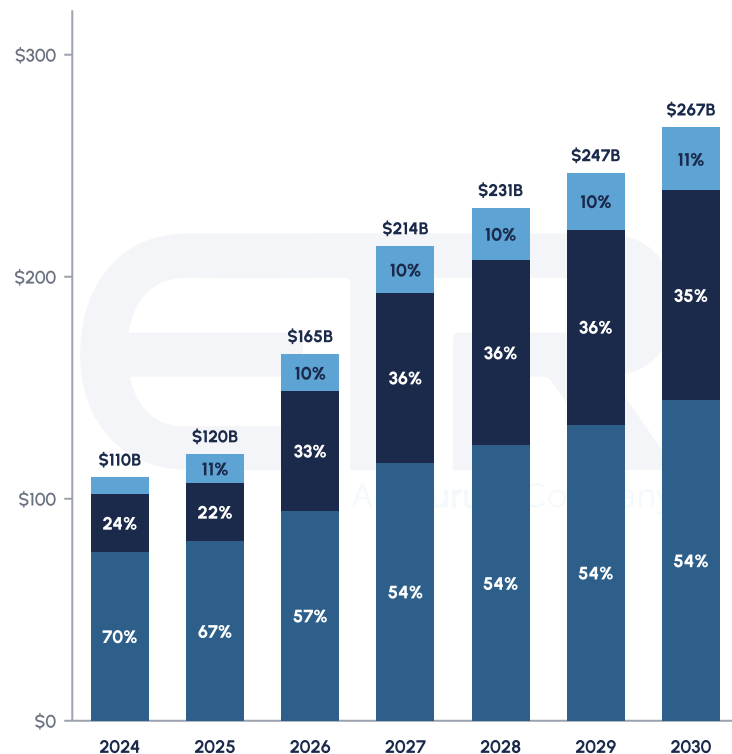
Source: ETR Core AI Model. AI-related WFE spending by segment: wafer capacity, capital intensity and required spend.

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Total WFE to trend to \$267-305B by 2030.

TOTAL WFE SPENDING BY SEGMENT (\$B)

Total industry spending: labels are share of total

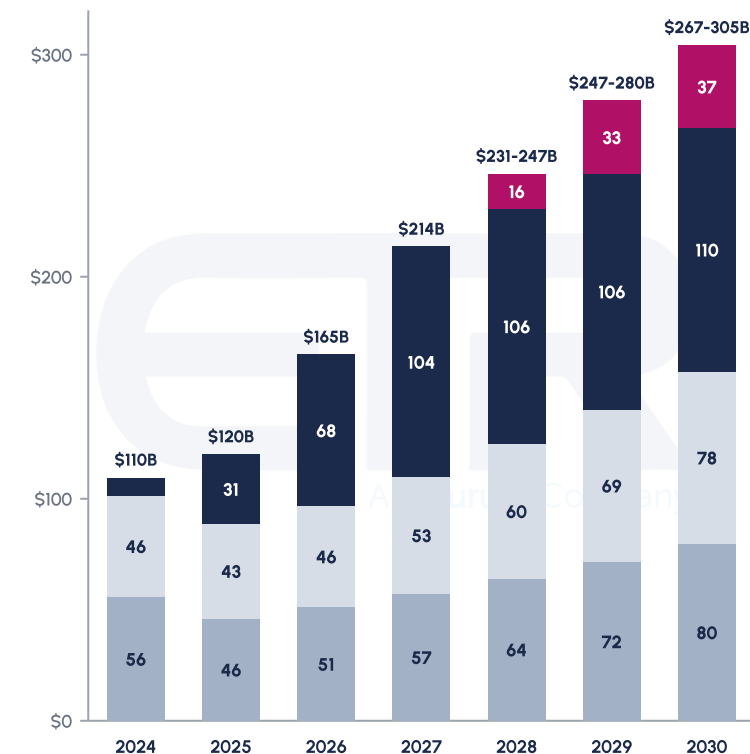


■ Logic and other ■ DRAM ■ NAND

Segment shares may not sum to 100 because of rounding.

Source: ETR Core AI Model. Total industry WFE spending by segment and by market.

TOTAL WFE BY MARKET (\$B)



■ Datacenter AI ■ Datacenter AI - upside

■ China ■ Rest of world ex. AI

OUR TAKE ON THE DATA

- **2030 base case:** AI datacenter WFE increases from **\$68B** this year to \$110B in 2030. Continued domestic Chinese spending and non-AI related WFE drives total WFE to **\$267B**.
- **Memory increasing in the mix:** from 33% last year to **46%** in 2030, primarily driven by DRAM.
- **Capacity absorption or continued build-out?**
Expected material growth beyond 2030 could front-load investments, with 2030 WFE trending to **over \$300B**.

WFE spending is set to more than double from 2025 to the end of the decade.

The sector is transitioning to multi-year agreements with price lock-ins.

LONG-TERM SUPPLY AGREEMENTS, 2026 DISCLOSURES

NVIDIA

"20-year leases to OpenAI [...] Each generation [...] could represent approximately **1.5 million NVIDIA GPUs**, or approximately **\$150 billion to \$200 billion in NVIDIA revenue**."

Colette Kress, EVP and CFO. 26 August 2026

SAMSUNG ELECTRONICS

"[...] DRAM and NAND supply volumes under multiyear supply contracts will easily account for **60% to 70% of our planned capacity**."

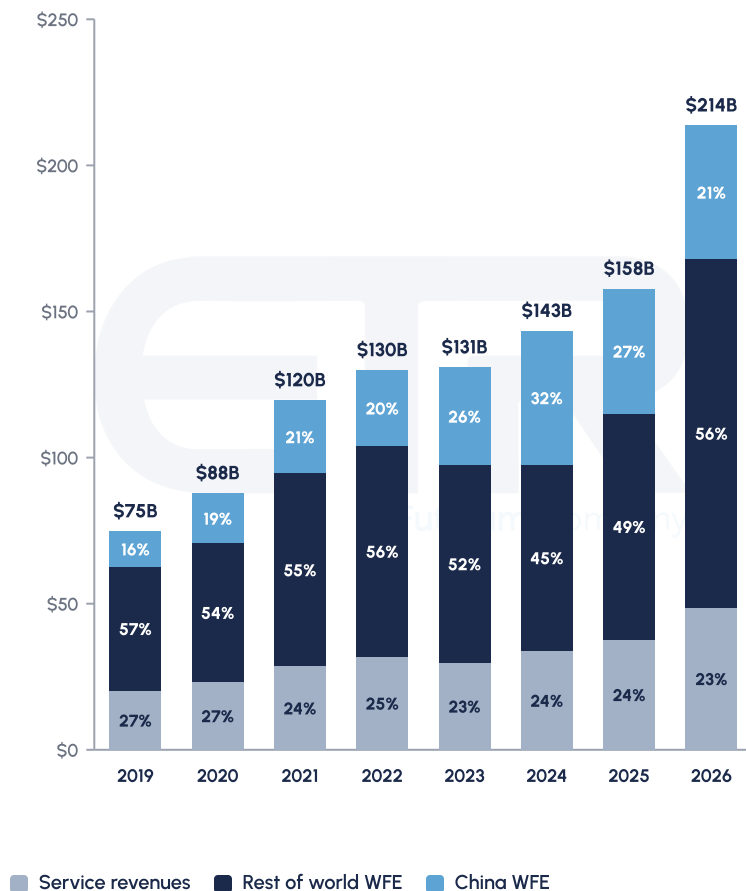
Jaejune Kim, EVP, Memory Division. 29 July 2026

MICRON

"[...] the **floor price (of long-term supply contracts) enables a very robust gross margin [...]** well above our peak quarterly margins in any past cycle."

Sanjay Mehrotra, Chairman, President and CEO. 24 June 2026

WAFER FAB EQUIPMENT AND SERVICE REVENUE (\$B)



OUR TAKE ON THE DATA

- **GPU and ASIC vendors lock in customer commitments years ahead:** Nvidia, Broadcom and AMD all focus on multi-year volume agreements to anchor their roadmap.
- **Memory vendors follow suit:** the leading memory vendors now have **50% to 70%** of capacity locked in under 3 to 5 year agreements with floor pricing.
- **Equipment sector remains largely capacity driven:** service revenue is **23%** of total sector revenue in 2026, down from 27% in 2019.

How can the SPE sector capitalize on the AI build-out while improving long-term resilience?

Source: ETR Core AI Model, company earnings calls and filings. Wafer fab equipment and service revenue, 2019 to 2026.

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